Source Regularity of the Rich6D Portfolio Economy

A rigorous check of the infinite Fourier target, the 128-term implementation, the frozen initial-CV rate, and the Sharpe–loss discrepancy

Research note for *The Law of Portfolio Learnability*

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Abstract

We distinguish the mathematical Rich6D population economy from its 128-term Fourier implementation. For the ideal infinite series, the population policy belongs to the six-dimensional Matérn RKHS $H^{9/2}$ but to no $H^{9/2+\varepsilon}$. The managed-return second-moment operator is uniformly comparable to the ordinary kernel covariance; an operator-range argument then proves that the largest admissible source order on the manuscript's interval $1 \le r \le 2$ is exactly $r = 1$. Finite truncation is smooth and its maximal *spectral* source order cannot be inferred from smoothness alone; nevertheless the 128-term truncation has a deterministic pointwise error below 7×10^{-9} in the rough loading map. We also show that selecting one regularization constant on a fixed finite initial-CV grid and freezing it retains the asymptotic $T^{-3/5}$ upper rate. Finally, an exact tangential–radial decomposition explains why response-one loss and out-of-sample Sharpe can move in different directions. The findings do not justify selecting a sample cutoff to reproduce a desired fitted exponent.

1 The population economy

Let $\Omega = (-1, 1)^6$, let K be the Matérn-3/2 kernel with fixed positive lengthscale on Ω, and let $\mathcal{H}_K$ denote its RKHS. The equivalent Sobolev order is

$$s = \nu + \frac{D}{2} = \frac{3}{2} + \frac{6}{2} = \frac{9}{2}, \qquad \mathcal{H}_K = H^{9/2}(\Omega) \tag{1}$$

with equivalent norms on the cube. All expectations below use the stationary joint law of characteristics and next-period returns.

For independent stationary groups $g = 1, \dots, G$, with $N = 3G$, write

$$U_{g,d,t} = \Phi(S_{g,d,t}), \quad S_{g,d,t+1} = \rho S_{g,d,t} + \sqrt{1-\rho^2}\,\eta_{g,d,t+1}, \quad Z_{g,k,d,t} = 2((U_{g,d,t} + k/3) \bmod 1) - 1, \tag{2}$$

for $d = 1, \dots, 6$ and $k = 0, 1, 2$. Here $|\rho| < 1$ and the innovations and stationary states are standard Gaussian. The marginal law of each stock's characteristic vector is uniform on Ω. Stocks within a triplet are dependent; they must not be modeled as independent individual draws.

Fix $\eta_0 = 0.35$ (to distinguish the nonlinearity amplitude from AR innovations). Define a period-two function

$$\psi_\infty(u) = c_\psi \sum_{m=2}^{\infty} \frac{\cos(\pi m u)}{m^5 \log(m+1)}, \qquad \psi_M(u) = c_\psi \sum_{m=2}^{M} \frac{\cos(\pi m u)}{m^5 \log(m+1)}. \tag{3}$$

The fixed normalization $c_\psi > 0$ is the same numerical Parseval normalization used by the implementation. Define

$$\theta(z) = \pi(z_1 + 1) + \eta_0 \sum_{d=2}^{6} \psi(z_d - z_1), \tag{4}$$

$$\beta(z) = \sqrt{c_\beta} \begin{pmatrix} \sqrt{2/3}\cos\theta(z) \\ \sqrt{2/3}\sin\theta(z) \\ 1/\sqrt{3} \end{pmatrix}, \qquad c_\beta = 0.04^2. \tag{5}$$

Both the ideal and truncated models use the same other economic parameters. Factor and idiosyncratic returns satisfy

$$R_{t+1} = B_t F_{t+1} + \varepsilon_{t+1}, \qquad F_{t+1} \sim N(\mu_F, I_3 - \mu_F \mu_F'), \qquad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2 I_N), \tag{6}$$

independently of the current characteristics, with $\sigma_\varepsilon = 0.08$ and $\|\mu_F\| < 1$. The rows of B_t are the corresponding $\beta(Z_{i,t})'$. In particular $\mathbb{E}[F_{t+1}F_{t+1}'] = I_3$.

Lemma 1.1 (Exact balanced economy and population policy)**.** *For every t, both with ψ_∞ and with ψ_M,*

$$\frac{B_t' B_t}{N} = g I_3, \qquad g := \frac{c_\beta}{3}. \tag{7}$$

The population-optimal characteristic policy for stock weights $W(Z_{i,t})/N$ is

$$W^\star(z) = \frac{\beta(z)'\mu_F}{g + \sigma_\varepsilon^2/N}. \tag{8}$$

Writing $q^\star = g\,\|\mu_F\|^2/(g + \sigma_\varepsilon^2/N)$, the optimal population Sharpe is $\mathrm{SR}^\star = \sqrt{q^\star/(1-q^\star)}$ in monthly units.

Proof. Under a triplet shift $k \mapsto k+1$, each difference $z_d - z_1$ changes by an integer multiple of two, so the period-two series in (4) is invariant. The leading phase rotates by $2\pi/3$. The three roots-of-unity identities therefore give $\sum_{k=0}^{2} \beta_{gk}\beta_{gk}' = c_\beta I_3$, proving (7). Conditional first and uncentered second moments of returns are $B_t\mu_F$ and $B_t B_t' + \sigma_\varepsilon^2 I_N$. The normal equation $\mathbb{E}[R_{t+1}R_{t+1}' \mid Z_t]w^* = \mathbb{E}[R_{t+1} \mid Z_t]$, the Woodbury identity, and (7) yield $w_t^* = (1/N)B_t\mu_F/(g + \sigma_\varepsilon^2/N)$. Its conditional mean and second moment are both $q^\star$, with variance $q^\star(1-q^\star)$. Since $0 < q^\star < 1$, the Sharpe formula follows. □

2 Spectral geometry and an exact source-space equivalence

Let $K_z = K(\cdot, z) \in \mathcal{H}_K$, and write

$$T_K := \mathbb{E}_{Z\sim\mathrm{Unif}(\Omega)}[K_Z \otimes K_Z], \qquad X_t := \frac{1}{N}\sum_{i=1}^{N} R_{i,t+1} K_{Z_{i,t}}, \qquad \Sigma := \mathbb{E}[X_t \otimes X_t]. \tag{9}$$

Here Σ is the *uncentered* managed-payoff second moment, not the centered return covariance.

Lemma 2.1 (Operator sandwich)**.** *For the exact infinite-series economy and for every finite Fourier truncation,*

$$\frac{\sigma_\varepsilon^2}{N} T_K \;\preceq\; \Sigma \;\preceq\; \left(c_\beta + \frac{\sigma_\varepsilon^2}{N}\right) T_K. \tag{10}$$

Consequently, for eigenvalues in decreasing order,

$$\mu_j(\Sigma) \asymp \mu_j(T_K) \asymp j^{-3/2}. \tag{11}$$

Moreover, for every $1 \le r \le 2$,

$$\mathrm{Ran}(\Sigma^{(r-1)/2}) = \mathrm{Ran}(T_K^{(r-1)/2}), \tag{12}$$

with equivalence of the corresponding minimal-preimage norms. No commutation or shared eigenvectors are required.

Proof. For any $h \in \mathcal{H}_K$, conditional on Z_t and using $\mathbb{E}[FF'] = I_3$,

$$\langle h, \Sigma h\rangle_{\mathcal{H}_K} = \mathbb{E}\left[\left\|\frac{1}{N}\sum_i h(Z_{i,t})\beta(Z_{i,t})\right\|_{\mathbb{R}^3}^2\right] + \frac{\sigma_\varepsilon^2}{N^2}\mathbb{E}\sum_i h(Z_{i,t})^2.$$

The first term is nonnegative and, by Cauchy–Schwarz and $\|\beta(z)\|^2 = c_\beta$, is bounded above by $c_\beta \mathbb{E}[N^{-1}\sum_i h(Z_{i,t})^2] = c_\beta \langle h, T_K h\rangle$. The noise term is exactly $(\sigma_\varepsilon^2/N)\langle h, T_K h\rangle$. This proves (10) without any cross-sectional independence assumption.

The Matérn-3/2 spectral density on $\mathbb{R}^6$ is proportional to $(\kappa^2 + \|\omega\|^2)^{-9/2}$. Standard Sobolev embedding/eigenvalue estimates on a bounded cube give $\mu_j(T_K) \asymp j^{-2(9/2)/6} = j^{-3/2}$; the min–max principle and (10) prove (11).

For (12), put $p = r - 1 \in [0, 1]$. The Löwner–Heinz inequality applied to (10) yields $c^p T_K^p \preceq \Sigma^p \preceq C^p T_K^p$ for the positive constants c, C in (10). Douglas' range-inclusion theorem, applied twice to $\Sigma^{p/2}$ and $T_K^{p/2}$, gives equality of the ranges and bounded factorizations in both directions. The claim at $p = 0$ is immediate. □

Lemma 2.2 (Necessary Sobolev regularity of a higher source)**.** *If* $W \in \mathcal{H}_K$ *satisfies*

$$W = \Sigma^{(r-1)/2} h \quad \text{for some } h \in \mathcal{H}_K, \qquad 1 \le r \le 2, \tag{13}$$

then, when identified with its scalar function on Ω,

$$W \in H^{(9/2)r}(\Omega). \tag{14}$$

The converse is not asserted: on a bounded nonperiodic domain, smoothness alone need not guarantee higher operator-range membership because of boundary conditions.

Proof. Let $J : \mathcal{H}_K \to L^2(\Omega)$ denote the canonical inclusion, $T_K = J^*J$ and $L_K = JJ^*$. The singular-value decomposition of J shows that $W \in \mathrm{Ran}(T_K^{(r-1)/2})$ if and only if $JW \in \mathrm{Ran}(L_K^{r/2})$. Indeed, if φ_j are orthonormal eigenfunctions of L_K with eigenvalues $\mu_j > 0$, both conditions are equivalent to

$$\sum_j \frac{|\langle JW, \varphi_j\rangle_{L^2}|^2}{\mu_j^r} < \infty. \tag{15}$$

Write $\mathfrak{H}_r := \mathrm{Ran}(L_K^{r/2})$ with its minimal-preimage norm. The canonical identity $\mathfrak{H}_1 = \mathcal{H}_K = H^{9/2}(\Omega)$ holds with equivalent norms. Moreover $L_K : L^2(\Omega) \to H^9(\Omega)$ continuously: extend its input by zero to $\mathbb{R}^6$, convolve with the Matérn kernel, and use its Fourier multiplier $(\kappa^2 + \|\omega\|^2)^{-9/2}$. Therefore $\mathfrak{H}_2 \hookrightarrow H^9(\Omega)$. Fractional powers of a positive injective operator

form an exact Hilbert interpolation scale, $\mathfrak{H}_r = [\mathfrak{H}_1, \mathfrak{H}_2]_{r-1}$ for $1 \le r \le 2$. By interpolation of continuous embeddings and the Sobolev interpolation identity on a cube, $\mathfrak{H}_r \hookrightarrow [H^{9/2}(\Omega), H^9(\Omega)]_{r-1} = H^{(9/2)r}(\Omega)$. The source-space equality (12) finishes the proof. □

3 Exact regularity of the infinite Fourier economy

Lemma 3.1 (The critical periodic series)**.** *The limiting series in* (3) *satisfies*

$$\psi_\infty \in H^{9/2}(\mathbb{R}/2\mathbb{Z}), \qquad \psi_\infty \notin H^{9/2+\delta}(\mathbb{R}/2\mathbb{Z}) \quad \text{for every } \delta > 0. \tag{16}$$

It is C^3, and its fourth derivative is unbounded near every point of $2\mathbb{Z}$.

Proof. By Fourier orthogonality, for $u \ge 0$,

$$\|\psi_\infty\|^2_{H^u(\mathbb{R}/2\mathbb{Z})} \asymp \sum_{m\ge 2} \frac{(1+m^2)^u}{m^{10}\log^2(m+1)} \asymp \sum_{m\ge 2} \frac{m^{2u-10}}{\log^2(m+1)}. \tag{17}$$

At $u = 9/2$ this is comparable to $\sum_{m\ge 2}(m\log^2 m)^{-1}$, which converges. At $u = 9/2+\delta$ it is comparable to $\sum m^{-1+2\delta}/\log^2 m$, which diverges. Since the series of absolute third-derivative coefficients is bounded by a constant times $\sum m^{-2}/\log m$, the function is C^3.

For $u \notin 2\mathbb{Z}$, the fourth derivative exists by Dirichlet summation and equals

$$\psi_\infty^{(4)}(u) = c_\psi \pi^4 \sum_{m\ge 2} \frac{\cos(\pi m u)}{m\log(m+1)}. \tag{18}$$

As $u \downarrow 0$, split the series at $M_u = \lfloor (3u)^{-1} \rfloor$. For $m \le M_u$, each cosine is at least $1/2$; this partial sum grows as $\frac{1}{2}\log\log(1/u)$. By summation by parts, the remaining signed tail is bounded by a constant times $[M_u \log(M_u+1)\, u]^{-1} = O(1/\log(1/u))$. Hence $\psi_\infty^{(4)}(u) \to +\infty$. Periodicity yields the other singular points. □

Theorem 3.2 (Exact maximal source order $r = 1$)**.** *Assume $\eta_0 \ne 0$ and $\mu_{F,1}^2 + \mu_{F,2}^2 > 0$, as in the frozen Rich6D economy. Let $W_\infty^\star$ be the population policy* (8) *constructed with the infinite series* (3)*. Then*

$$W_\infty^\star \in \mathcal{H}_K, \qquad W_\infty^\star \notin \operatorname{Ran}(\Sigma_\infty^{(r-1)/2}) \quad \text{for every } r \in (1, 2]. \tag{19}$$

Thus the maximal source order on the paper's interval $1 \le r \le 2$ is exactly $r = 1$. This conclusion does not require that Σ_∞ and T_K have common eigenfunctions.

Proof. Membership. The pullbacks $z \mapsto \psi_\infty(z_d - z_1)$ belong to $H^{9/2}(\Omega)$ by the elementary Sobolev pullback theorem for the linear rank-one submersions $z \mapsto z_d - z_1$ on a bounded cube. Since $9/2 > D/2 = 3$, $H^{9/2}(\Omega)$ is an algebra and is stable under smooth composition on bounded ranges. Equations (4)–(8) therefore yield $W_\infty^\star \in H^{9/2}(\Omega) = \mathcal{H}_K$.

Failure of every higher Sobolev order. Fix $z_1, z_3, \dots, z_6$ and vary z_2 near $z_2 = z_1$; write $u = z_2 - z_1$. Apart from a constant term, (8) has the form

$$A\cos\big(\theta_0 + \eta_0\psi_\infty(u)\big) + B\sin\big(\theta_0 + \eta_0\psi_\infty(u)\big), \tag{20}$$

where $(A, B) \neq (0, 0)$, and θ_0 depends continuously on the fixed coordinates. By the chain rule on $u \neq 0$, its fourth derivative in z_2 is

$$\eta_0[-A\sin\theta + B\cos\theta]\,\psi_\infty^{(4)}(u) + \text{terms bounded near } u = 0. \tag{21}$$

The remainder is bounded because $\psi_\infty \in C^3$. The bracket tends to a continuous nonzero value for an open, positive-measure set of fixed coordinates: choose $z_3 = \cdots = z_6 = z_1$, then $\theta = \pi(z_1 + 1) + 5\eta_0\psi_\infty(0)$ at $u = 0$, and vary z_1; the nontrivial sinusoidal bracket cannot vanish identically. For this positive-measure set of one-dimensional slices, (21) is unbounded near an interior point, by Lemma 3.1.

If $W_\infty^\star \in H^{9/2+\delta}(\Omega)$ for any $\delta > 0$, an extension to $H^{9/2+\delta}(\mathbb{R}^6)$ and Fubini/Plancherel imply that almost every z_2-slice belongs locally to $H^{9/2+\delta}(\mathbb{R})$. In one dimension, $H^{9/2+\delta}$ embeds continuously into C^4 on compact intervals, contradicting the unbounded fourth derivative on the positive-measure collection of slices. Hence $W_\infty^\star \notin H^{9/2+\delta}(\Omega)$ for every $\delta > 0$.

Spectral-source conclusion. If the source condition held for any $r > 1$, Lemma 2.2 would give $W_\infty^\star \in H^{(9/2)r}(\Omega) \subset H^{9/2+\delta}(\Omega)$ for $\delta = (9/2)(r-1) > 0$, a contradiction. □

Remark 3.3 (Scope of the result). Theorem 3.2 establishes source-order sharpness for the *ideal infinite* economic target. It is not an identification of a minimax lower bound for a single economy. In particular, it does not imply that an ordinary finite-horizon log–log regression must have slope $-3/5$.

4 What finite Fourier truncation does and does not establish

Proposition 4.1 (Deterministic truncation and nonuniform higher smoothness). *For $M \geq 2$,*

$$\|\psi_\infty - \psi_M\|_\infty \leq \frac{c_\psi}{4M^4\log(M+1)}. \tag{22}$$

At $M = 128$, using the frozen numerical normalization $c_\psi \approx 34.95703934$, this upper bound is 6.6991×10^{-9}. The corresponding uniform bounds are

$$\|\beta_\infty(z) - \beta_M(z)\|_{\mathbb{R}^3} \leq \sqrt{\frac{2c_\beta}{3}}\,5|\eta_0|\;\|\psi_\infty - \psi_M\|_\infty \leq 3.829 \times 10^{-10}, \tag{23}$$

$$\|W_\infty^\star - W_M^\star\|_\infty \leq \frac{\|\mu_F\|}{g + \sigma_\varepsilon^2/N}\sqrt{\frac{2c_\beta}{3}}\,5|\eta_0|\;\|\psi_\infty - \psi_M\|_\infty. \tag{24}$$

In the $N = 600$ calibrated experiment the last bound is 4.654×10^{-7} for scalar $W^\star(z)$, or 7.756×10^{-10} per stock after division by N.

Every finite $W_M^\star$ is C^∞ on the closed cube and satisfies the $r = 1$ source condition. The theorem does not *claim that $W_M^\star$ has a specific maximal $r > 1$: this requires checking the Matérn operator-range and boundary conditions. Nevertheless, for every fixed $r > 1$, the higher-source preimage norms of $W_M^\star$ (when defined) cannot remain bounded uniformly as $M \to \infty$.*

Proof. Since $m \mapsto m^{-5}/\log(m+1)$ is decreasing, $\sum_{m>M} m^{-5}/\log(m+1) \leq \int_M^\infty x^{-5}/\log(x+1)\,dx \leq [4M^4\log(M+1)]^{-1}$. This proves (22). The cosine–sine vector in (5) is globally Lipschitz in its phase with constant $\sqrt{2c_\beta/3}$; the phase changes by at most $5|\eta_0|\,\|\psi_\infty - \psi_M\|_\infty$.

Multiplying by $\|\mu_F\|/(g+\sigma_\varepsilon^2/N)$ proves the remaining bounds. All displayed numerical bounds are direct substitutions, not Monte Carlo estimates.

Each ψ_M is a trigonometric polynomial, so $W_M^\star$ is smooth. The Fourier calculation in Lemma 3.1 also implies $\psi_M \to \psi_\infty$ in $H^{9/2}$, and continuity of smooth composition in this Sobolev algebra gives $W_M^\star \to W_\infty^\star$ in $\mathcal{H}_K$.

Fix $r > 1$ and put $\alpha = (r-1)/2$. By Lemma 2.1, the source space for each Σ_M equals the *same* $\mathrm{Ran}(T_K^\alpha)$ with uniformly comparable source norms. Suppose for contradiction that $W_M^\star = T_K^\alpha h_M$ with $\sup_M \|h_M\|_{\mathcal{H}_K} < \infty$. Extract a weakly convergent subsequence $h_{M_k} \rightharpoonup h$ in $\mathcal{H}_K$. Then $W_{M_k}^\star = T_K^\alpha h_{M_k} \rightharpoonup T_K^\alpha h$, while $W_{M_k}^\star \to W_\infty^\star$ strongly in $\mathcal{H}_K$. Hence $W_\infty^\star = T_K^\alpha h$, contradicting Theorem 3.2 and (12). Thus no uniform higher-source bound exists. □

Remark 4.2 (The norms tell a different story from pointwise error). For every fixed $\delta > 0$, the periodic Sobolev calculation gives

$$\|\psi_M\|_{H^{9/2+\delta}}^2 \asymp \sum_{m \le M} \frac{m^{-1+2\delta}}{\log^2(m+1)} \asymp \frac{M^{2\delta}}{2\delta \log^2 M} \quad (M \to \infty).$$

Thus a tiny pointwise truncation error does not imply accurate approximation of higher source norms. The current simulation cannot numerically certify the maximal source order from 128 Fourier terms; the ideal economy, however, has a rigorous $r = 1$ proof.

5 A constant calibrated once by initial CV still has the theoretical rate

Proposition 5.1 (Frozen initial-CV preservation of the upper rate)**.** *Suppose the manuscript's stationary, mixing, tail, spectral and $r = 1$ conditions hold, with $b = 3/2$. Let $\mathcal{A} = \{a_1, \dots, a_J\} \subset (0, \infty)$ be a fixed* finite *set, independent of T, and let $\widehat{a}_{T_0} \in \mathcal{A}$ be any measurable selection from the first T_0 observations, with fixed T_0 (in the simulation $T_0 = 60$, $J = 71$). For each $T \ge T_0$ fit the entire expanding training history with*

$$\lambda_T = \widehat{a}_{T_0} T^{-3/5}.$$

Then, as $T \to \infty$,

$$Q(\widehat{W}_{\lambda_T,T}) - Q(W^\star) = O_{\mathbb{P}}(T^{-3/5}), \qquad \mathrm{SR}^\star - \mathrm{SR}(\widehat{W}_{\lambda_T,T}) = O_{\mathbb{P}}(T^{-3/5}). \tag{25}$$

The selection may be correlated with the eventual training history. No initial-CV consistency or independence of the selected constant is needed for these rate upper bounds.

Proof. For every deterministic $a_j \in \mathcal{A}$, the manuscript's $r = 1$ bound, together with $\mathcal{C}(\lambda) \lesssim \lambda^{-2/3}$, gives for sufficiently large T

$$Q(\widehat{W}_{a_j T^{-3/5},T}) - Q(W^\star) \le \left(9R^2 a_j + C_{\delta,J} a_j^{-2/3}\right) T^{-3/5} \tag{26}$$

with probability at least $1 - \delta/J$. Here the admissibility threshold can depend on the positive constant a_j and δ/J, and $C_{\delta,J}$ includes the common spectral constant. Since $J < \infty$, take the maximum of these finitely many constants and the maximum of their finite asymptotic thresholds. A union bound makes (26) hold simultaneously for all j with probability at least

$1-\delta$. It must then also hold for the data-selected index. The manuscript's loss-to-Sharpe lemma transfers the same order to positive population Sharpe. The selection does not enter a conditional independence step. □

Remark 5.2 (Why finite log–log slopes need not equal $-3/5$). The rate is a big-$O_{\mathbb{P}}$ upper bound, not a two-sided expansion. Its constants include a and $a^{-2/3}$; a small initial choice can have large finite-horizon constants. For $a \in [10^{-8}, 10^{-1}]$, the lower endpoint gives $a^{-2/3} \approx 2.15\times10^{5}$. Neither the theorem nor the CV procedure requires the observed slope over $60 \le T \le 4860$ to be -0.6.

6 Why lower response-one regret need not mean higher Sharpe

Define the population inner product $\langle f, g\rangle_\Sigma := \langle f, \Sigma g\rangle_{\mathcal{H}_K}$ and let $q = \|W^\star\|_\Sigma^2 \in (0,1)$. For any policy W with positive population expected payoff set

$$c := \frac{\langle W, W^\star\rangle_\Sigma}{q} > 0, \qquad U := W - cW^\star, \qquad \langle U, W^\star\rangle_\Sigma = 0, \qquad d := \|U\|_\Sigma^2 . \tag{27}$$

Proposition 6.1 (Exact radial/angular portfolio decomposition)**.** *Under the population normal equation $\Sigma W^\star = \mathbb{E}X_t$,*

$$Q(W) - Q(W^\star) = q(c-1)^2 + d, \tag{28}$$

$$\mathrm{SR}(W) = \frac{cq}{\sqrt{c^2 q(1-q) + d}} = \frac{\mathrm{SR}^\star}{\sqrt{1 + d/[c^2 q(1-q)]}}. \tag{29}$$

Hence the response-one objective penalizes radial misscaling and angular error, while Sharpe is invariant to any positive radial rescaling of a fixed direction. In particular, $Q(cW^\star) - Q(W^\star) = q(c-1)^2$ but $\mathrm{SR}(cW^\star) = \mathrm{SR}^\star$ for every $c > 0$.

Proof. The population normal equation gives $\mathbb{E}R^p(W) = \langle W, \Sigma W^\star\rangle = cq$. The uncentered second moment is $\mathbb{E}[R^p(W)^2] = \|W\|_\Sigma^2 = c^2 q + d$. Therefore the payoff variance is $c^2 q(1-q) + d$. The Sharpe identity follows directly. Expanding the quadratic criterion around its optimum yields $Q(W) - Q(W^\star) = \|W - W^\star\|_\Sigma^2 = q(c-1)^2 + d$. □

This identity offers a direct interpretation of the archived baseline results at $T = 4860$: the initially CV-calibrated policy has mean quadratic regret about 0.0113 versus about 0.0222 under the old rule, but mean annual Sharpe about 2.9325 versus 2.9348. More directions may improve radial response-one fit and simultaneously degrade the learned tangency *direction*. The losses and Sharpe ratios are therefore not interchangeable diagnostics.

If the target is explicitly Sharpe, a scale-invariant population criterion is

$$Q_{\mathrm{ray}}(W) := \min_{c \ge 0} \mathbb{E}[(1 - cR^p(W))^2] = 1 - \frac{\left(\mathbb{E}R^p(W)\right)_+^2}{\mathbb{E}R^p(W)^2}, \tag{30}$$

which, for positive mean, equals $1/[1 + \mathrm{SR}(W)^2]$. A corresponding initial validation score is possible, but would be *a different tuning protocol* and needs its own locked independent evaluation. With only 40 pooled initial validation payoffs, noisy Sharpe selection is a serious finite-sample consideration.

7 Theoretical conclusions and recommended presentation

1. The *ideal* Rich6D economy satisfies $b = 3/2$ and has exactly $r = 1$ on the paper's source interval. Therefore the theory permits the benchmark rates $\lambda_T \asymp T^{-3/5}$, $\mathcal{C}(\lambda_T) \asymp T^{2/5}$ and the upper bound $Q(\widehat{W}) - Q(W^\star) = O_{\mathbb{P}}(T^{-3/5})$, with the paper's dependence/tail assumptions and sufficient asymptotic stability conditions.

2. The finite $M = 128$ simulated model is not itself proved to have *maximal* source order one, although it approximates the ideal target extremely well pointwise. The largest finite-sample power-law exponent observed in a fit cannot identify maximal operator smoothness.

3. Initial-CV calibration on a predetermined finite grid preserves the asymptotic $T^{-3/5}$ upper rate. The archived CV full-window Sharpe-gap slope near -0.399 is not a mathematical contradiction of that statement.

4. The main financial mechanism behind different response-one and Sharpe outcomes is the exact radial/angular decomposition (28)–(29). The researcher should not substitute response-one regret for Sharpe-directional error in interpreting CV.

5. Do not select $T_{\max}$ after observing which interval makes a fitted slope closest to -0.6. A defensible main figure can use a predetermined economic horizon; provide the full grid and sensitivity windows elsewhere, labeling all fitted exponents as descriptive, not empirical proofs of a minimax theorem.

6. Numerically unresolved rank-4096/8192 projection floors remain explicit caveats; proving source regularity analytically does not certify the finite-rank approximation.

Documentation checked. The portfolio-learnability repository contains the Rich6D source, the theoretical assumptions, the paired rank-8192 audit, and the completed initial-CV results. Those sources were inspected on 11 October 2026. This note gives an independent mathematical derivation; it is not a claim that every simulation or proof in the manuscript has been independently rerun.

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